

Reading and Running AI-Generated Code

CS1 Week 3 — Wednesday, September 2, 2026

Computer Science I

The lesson in one sentence

A chat AI tool gives you a fast first draft; it only knows what you told it. Read what it wrote, notice the gaps, and verify.

Predict. Read the code. Try one small change. Verify.

Today's example: the World Bible

- CS1 is deliberately freeform — no textbook, four possible starter “worlds” to build from.
- Today's world: a frontier settlement with a food stock, a settler count, and a warning rule.
- The rule in plain English: *if the food stock is below a threshold, print a warning.*

From class: a plain-English rule turned into a first-draft Python script

Work IQ GPT 5.6 Think

particular real group.

frontier_settlement.py

```
10 print( ***** )
11
12 settlement_name = "Dusty Trail"
13 number_of_settlers = 12
14 food_stock = 8
15
16 print()
17 print(f"Howdy, traveler! Welcome to {settlement_name}.")
18 print(f"We've got {number_of_settlers} settlers in town.")
19 print(f"The storehouse currently holds {food_stock} crates of food.")
20
21 # -----
22 # PART 1: THE ASSIGNMENT'S ONE IF STATEMENT
23 # -----
```

+ Message Copilot

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AI-generated content may be incorrect

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What the tool did — and didn't — know

- The prompt only said: check a threshold, print a warning, use a “settler” voice.
- It did **not** say where to save the file or how to run it.
- That's expected: those are things you're expected to notice and supply yourself.

Reading the generated code

- `#` comment lines — notes for humans, ignored by Python.
- `print(...)` — sends text to the screen.
- `settlement_name = "Dusty Trail"` — a variable: a name that gives a bare value context.

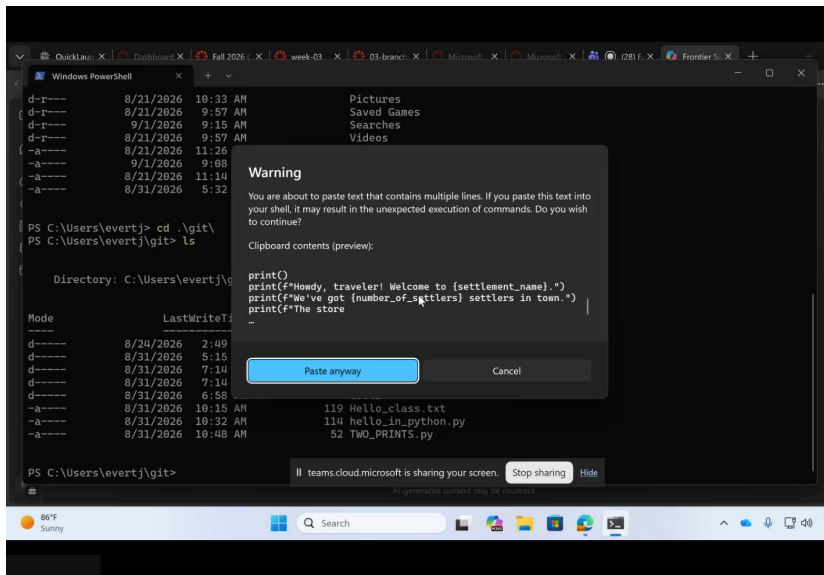
A number alone (`"8"`) means nothing. `food_stock = 8` means something.

Moving the code into a file: here-string + pipe

```
$code = @'
# generated Python code goes here
'@
$code | Set-Content -Path frontiersettlement.py -Encoding utf8
```

- `@' ... '@` is a **here-string**: a block of multi-line text.
- `|` is the **pipe**: what goes in one end comes out the other, unchanged.
- `Set-Content` writes the piped text into a file.

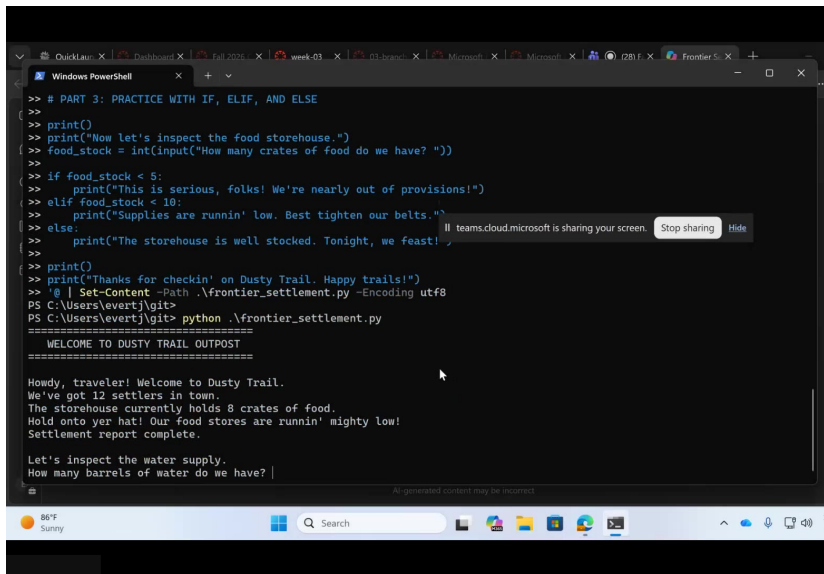
From class: PowerShell's multi-line-paste warning — expected, not an error



Reading the warning

- PowerShell is confirming you meant to run *all* of the pasted text, not just the first line.
- Stray characters — an extra quote, an extra space, a smart quote or emoji some tools insert — can break a paste like this.
- If a pasted block looks off, reread it before choosing “Paste anyway.”

From class: running frontiersettlement.py and reading its output



```
>> # PART 3: PRACTICE WITH IF, ELIF, AND ELSE
>>
>> print()
>> print("Now let's inspect the food storehouse.")
>> food_stock = int(input("How many crates of food do we have? "))
>>
>> if food_stock < 5:
>>     print("This is serious, folks! We're nearly out of provisions!")
>> elif food_stock < 10:
>>     print("Supplies are runnin' low. Best tighten our belts.")
>> else:
>>     print("The storehouse is well stocked. Tonight, we feast!")
>>
>> print()
>> print("Thanks for checkin' on Dusty Trail. Happy trails!")
>> '@ | Set-Content -Path .\frontier_settlement.py -Encoding utf8
PS C:\Users\evvertj\git>
PS C:\Users\evvertj\git> python .\frontier_settlement.py
=====
WELCOME TO DUSTY TRAIL OUTPOST
=====

Howdy, traveler! Welcome to Dusty Trail.
We've got 12 settlers in town.
The storehouse currently holds 8 crates of food.
Hold onto yer hat! Our food stores are runnin' mighty low!
Settlement report complete.

Let's inspect the water supply.
How many barrels of water do we have? |
```

From class: the same file, opened in Notepad — still just text on disk

```
File Edit View
# PART 1: THE ASSIGNMENT'S ONE IF STATEMENT

if food_stock < 10:
    print("Hold onto yer hat! Our food stores are runnin' mighty low!")

print("Settlement report complete.")
# PART 2: PRACTICE WITH IF AND ELSE

print()
print("Let's inspect the water supply.")
water_barrels = int(input("How many barrels of water do we have? "))

if water_barrels >= 10:
    print("Plenty of water, partner. The settlement is in fine shape!")
else:
    print("Well, butter my biscuit. We'd better find water before sundown!")

# PART 3: PRACTICE WITH IF, ELIF, AND ELSE

print()
print("Now let's inspect the food storehouse.")
food_stock = int(input("How many crates of food do we have? "))

if food_stock < 5:
    print("This is serious, folks! We're nearly out of provisions!")
elif food_stock < 10:
    print("Supplies are runnin' low. Best tighten our belts.")
else:
    print("The storehouse is well stocked. Tonight, we feast!")

print()
print("Thanks for checkin' on Dusty Trail. Happy trails!")
```

Ln 28, Col 1 29 of 1,531 characters Plain text 100% Unix (LF) UTF-8 w

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The if block, read line by line

```
if food_stock < 10:  
    print("Hold onto yer hat! Our food stores are runnin' mighty low!")
```

- An if block runs only when its condition is true.
- Study this shape, then copy one line and change it to see what happens.

Tool posture

- If something is a mystery, confusing, or seems wrong: ask the tool to explain it.
- You stay the one deciding whether the explanation makes sense.
- The tool is only guessing at what will help you — it can't read your mind.

Paired reasoning starts today

- **Foreman:** plans on paper, doesn't type.
- **Worker:** has the keyboard, executes the plan.
- If you were a worker last week, try being the foreman this week.
- Keep the same partner from last week where you can.

Before Friday

- Continue the paired work started in class.
- Be ready to explain what each variable represents.
- Be ready to explain what the `if` condition is checking.

Read before you run. Ask when it's unclear. You decide what makes sense.